Muscle Activation Patterns During Different Squat Techniques

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Abstract

Slater, LV, and Hart, JM. Muscle activation patterns during different squat techniques. *J Strength Cond Res* 31(3): 667–676, 2017—Bilateral squats are frequently used exercises in sport performance programs. Lower extremity muscle activation may change based on knee alignment during the performance of the exercise. The purpose of this study was to compare lower extremity muscle activation patterns during different squat techniques. Twenty-eight healthy, uninjured subjects (19 women, 9 men, 21.5 ± 3 years, 170 ± 8.4 cm, 65.7 ± 11.8 kg) volunteered. Electromyography (EMG) electrodes were placed on the vastus lateralis, vastus medialis, rectus femoris, biceps femoris, and the gastrocnemius of the dominant leg. Participants completed 5 squats while purposefully displacing the knee anteriorly (AP malaligned), 5 squats while purposefully displacing the knee medially (ML malaligned) and 5 squats with control alignment (control). Normalized EMG data (MVIC) were reduced to 100 points and represented as percentage of squat cycle with 50% representing peak knee flexion and 0 and 99% representing fully extended. Vastus lateralis, medialis, and rectus femoris activity decreased in the medio-lateral (ML) malaligned squat compared with the control squat. In the antero-posterior (AP) malaligned squat, the vastus lateralis, medialis, and rectus femoris activity decreased during initial descent and final ascent; however, vastus lateralis and rectus femoris activation increased during initial ascent compared with the control squat. The biceps femoris and gastrocnemius displayed increased activation during both malaligned squats compared with the control squat. In conclusion, participants had altered muscle activation patterns during squats with intentional frontal and sagittal malalignment as demonstrated by changes in quadriceps, biceps femoris, and gastrocnemius activation during the squat cycle.

Key Words quadriceps, knee, performance, rehabilitation

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Introduction

Bilateral squats are a staple exercise in most sport performance and knee rehabilitation programs. Despite its popularity in gyms and sports medicine clinics, there is little research on muscle activation patterns during an unloaded bodyweight bilateral squat other than its use to strengthen the quadriceps. Previous researchers (4,18,24) have noted high quadriceps activation and little hamstring activation during the descending, holding, and ascending phases of the squat, supporting the use of the bilateral squat for quadriceps strengthening in rehabilitation and performance programs.

Although the squat is a widely accepted exercise to strengthen the thigh musculature, sports medicine and performance professionals teach a variety of techniques, most commonly changing the stance width and depth of the squat. Foot abduction driven by hip rotation and stance width generally vary among practitioners and practice, however no significant difference in quadriceps muscle activation patterns have been noted when comparing narrow and wide stance and varying foot positions (12,32). However, increased adductor longus and gluteus maximus activity during a wide stance squat have been reported (32). This suggests that different stance widths do not change the use of the squat as a quadriceps strengthening exercise, however they may help target adjacent muscles. Another squat technique variation, the deep squat where maximal knee flexion is encouraged, may result in increased gluteus maximus activation during the ascending phase of the squat (4), however increased squat depth using relative loads may not increase gluteal activation (6). Although the full squat may not increase hip involvement, poorly performed squats have been associated with altered gluteal activation (7), indicating that changes in squat performance may alter muscle involvement.

A poorly performed squat may result in altered lower extremity alignment such as increased knee valgus which may expose the lower extremity joints to excessive torques that may require adaptive muscle activation strategies to stabilize the lower extremity joints. Although many sports medicine and performance professionals are comfortable instructing patients to execute proper squats, there is little information regarding differences in muscle activation patterns in the lower extremity muscles during squats with

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varying alignments. Furthermore, strength and conditioning coaches often design client programs based on performance on functional screenings and assessments, including the bilateral and single-leg squat (2,7,20). Understanding if different lower extremity alignments during a squat change muscle activation patterns in the lower extremity will provide an evidence-based approach to coaching patients on appropriate squat alignment and designing effective strengthening programs.

Consideration for lower extremity alignment during the bilateral squat is also important because of the potential for increased patellofemoral contact forces during knee flexion (3,33,39,41). Some models have predicted peak force during the squat to be around 90–100° of knee flexion (14,15), which is common during squat exercises. Because the knee deviates from neutral alignment near peak knee flexion, different patterns of muscle activation may be necessary to attenuate the increased patellofemoral forces and stabilize the knee joint. For example, decreased vastus lateralis and increased gastrocnemius muscle activation have been reported during squats with medial knee displacement compared with a neutrally aligned squat (29,36). However, little is known about the muscle activation patterns in the rectus femoris and knee flexors during knee joint deviations while squatting. Increased knee flexor activation during bilateral squats may increase ligamentous strain to stabilize the knee joint (37). Therefore, bilateral squat positions that increase muscle activation in the hamstrings may increase knee injury risk. This is particularly important given the growing popularity of the ballet plié squat where clients purposefully lift their heels off the ground and squat with weight at their toes despite a lack of information about the way the lower extremity musculature stabilizes the knee joint during the increased anterior displacement. Therefore, the purpose of this study was to compare lower extremity electromyographic muscle activation during a neutrally aligned squat compared with antero-posterior (AP) malaligned and medio-lateral (ML) malaligned bilateral squats. We hypothesized that malaligned squats would result in increased quadriceps, hamstring, and gastrocnemii activity compared with control squats.

Methods

Experimental Approach to the Problem

A descriptive, repeated measures laboratory study was used to compare muscle activation patterns during the control, AP malaligned, and ML malaligned bilateral squats. The experimental approach provided unique information about the muscle activation patterns during each squat technique to assist sports medicine and performance professionals with information about differences in lower extremity muscle activation patterns and strategies during commonly performed malaligned squats. The independent variable in this study was the squat technique (control, AP and ML aligned squats). The dependent variables were lower extremity muscle activation pattern during the squat cycle measured with surface electromyography.

Subjects

Twenty-eight healthy, recreationally active participants (19 women, 9 men) without self-reported history of lower extremity injury volunteered (21.5 ± 3 years, 170 ± 8.4 cm, 65.7 ± 11.8 kg). All participants were familiar with the squat exercise. Exclusion criteria included history of lower extremity injury within previous 6 months, history of low back pain or lower extremity joint surgery, pregnancy, known muscular abnormalities, and known degenerative joint disease. All participants signed informed consent approved by the university's institutional review board.

Instrumentation

A wireless surface electromyography (EMG) system (Trigno Sensor System, Delsys Inc., Natick, MA, USA: interelectrode distance = 10 mm, 80 dB common mode rejection rate) was used to record lower extremity muscle activity. Electromyography data were sampled at 2,000 Hz. Maximal voluntary isometric contractions were exported using EMGworks Analysis software (version 4.1.1.0; Delsys Inc.). An electromagnetic motion-analysis system (Ascension Technology Corporation, Burlington, VT, USA) was used during collection. Kinematic data were sampled at 144 Hz. Three-dimensional joint angles and EMG data were synchronized, reduced, and exported using MotionMonitor software (Innovative Sports Training, Chicago, IL, USA).

Electromyography Electrode Placement

The electrodes for the quadriceps muscles were placed on the distal third of the participant's vastus lateralis and vastus medialis and the proximal third of the participant's rectus femoris. The lateral and medial gastrocnemius electrodes were placed at 20% of the distance of the shank from the knee joint line to the lateral malleolus (36). The electrode on the biceps femoris was placed halfway between the ischial tuberosity and the lateral epicondyle of the tibia (19).

Procedures

Participants reported to the laboratory for a single session wearing athletic shoes and athletic clothing. Electromyography electrodes were placed over the muscles of interest on the participant's dominant leg, defined as the preferred kicking leg, after the skin was shaved, lightly abraded, and cleaned with alcohol. After electromagnetic sensors were attached, participants placed the dominant leg within the boundaries of a single force plate embedded in the floor and the contralateral leg on the floor, outside of the force plate (13) (Figure 1). The participant practiced bilateral squats to parallel to become accustomed to the wires from the electromagnetic motion capture system. The participant was asked to stand with feet shoulder width apart, toes pointing forward and was instructed to perform 5 squats to 90° of flexion with knees collapsing inward (ML malaligned), 5 squats to 90° of flexion while lifting heels off the floor (AP

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Figure 1. Participants performed 5 medio-lateral malaligned squats (A, D) followed by 5 antero-posterior malaligned squats (B, E) followed by 5 control squats (C, F). Participants rested for 1 minute between each squat repetition. No feedback was provided during any of the squat techniques other than the control squat.

malaligned), and 5 squats to 90° of flexion while keeping heels on the floor and knees in line with feet (control) (Figure 1). Feedback was only given during the control squat and was standardized to include the following statements: Sit back at your heels like you're sitting in a chair; push your knees out in the bottom of the squat; keep your toes pointing forward.

Normalization Procedures

Maximal voluntary isometric contractions (MVICs) were collected before the participant completed any squats. Maximal voluntary isometric contractions for the vastus lateralis, vastus medialis, rectus femoris, and biceps femoris were collected in short sitting with the knees flexed to 90° using a gait belt around the distal third of the shank during both isometric knee extension and knee flexion. Ninety degrees was used to normalize quadriceps and hamstring activation to maximal activity during peak knee flexion. Maximal voluntary isometric contractions for the lateral and medial gastrocnemius were collected with the subject lying prone and 10° of plantarflexion. Knee flexion and ankle plantarflexion were measured using a goniometer. Three 5-second MVIC trials were collected in each position, averaging the middle 3 seconds of each trial for the individual muscles. All muscle activity was normalized and expressed as a percentage of MVIC.

Statistical Analyses

The raw EMG data were filtered and exported using the MotionMonitor software, utilizing a bandpass filter (10–450 Hz) with a 60 Hz notch filter and a 50 milliseconds window, moving average, root mean square algorithm. The EMG and kinematic data were synchronized and reduced to 100 points to represent 100% of the squat cycle, where 50% represents peak knee flexion and 0 and 99% represent full knee extension (27). Initial and final descent were defined as 0–24 and 25–49%, respectively. Initial and final ascent were defined as

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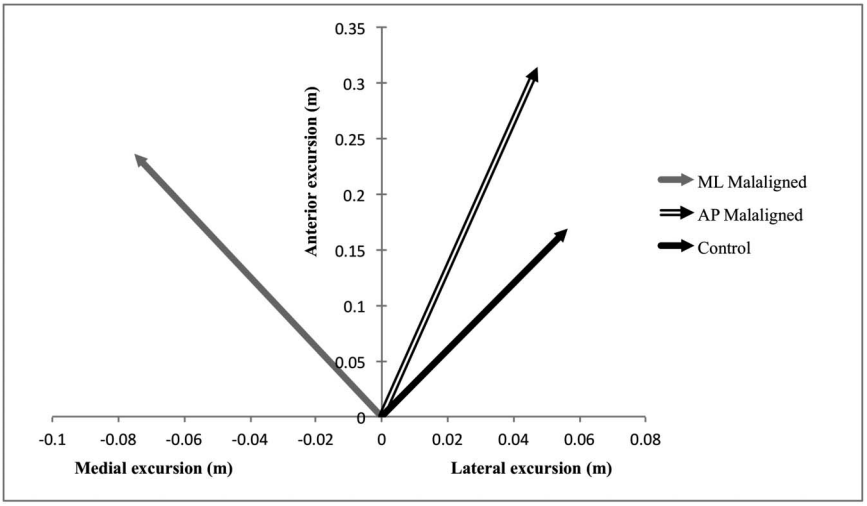


Figure 2. Peak knee joint excursion from full knee extension at the beginning of the squat.

50–74 and 75–99%, respectively. After being reduced to 100 points, data were smoothed using a 3-point moving average window and 90% confidence intervals were calculated about the mean of each percentage point. Means and 90% confidence intervals were calculated for each muscle during each squat technique. Areas in which the confidence intervals did not overlap for more than 3 consecutive percentage points were considered statistically significant (9,21). Mean differences and associated pooled standard deviations were calculated for each muscle during periods of the squat cycle when squat techniques were significantly different. Cohen's *d* effect sizes using mean differences and pooled standard deviations were calculated for each muscle. Effect sizes were interpreted as weak (<0.2), small (0.21–0.39), moderate (0.4–0.7), large (0.71–0.99), and very large (>1.0).

Results

Medio-Lateral Malaligned Squat

Participants demonstrated increased anterior and medial knee displacement compared with the control squat (Figure 2). The ML malaligned squat resulted in significantly increased dorsiflexion, ankle inversion, knee flexion, knee abduction, and hip adduction during approximately 10–85% of the squat cycle compared with the control squat.

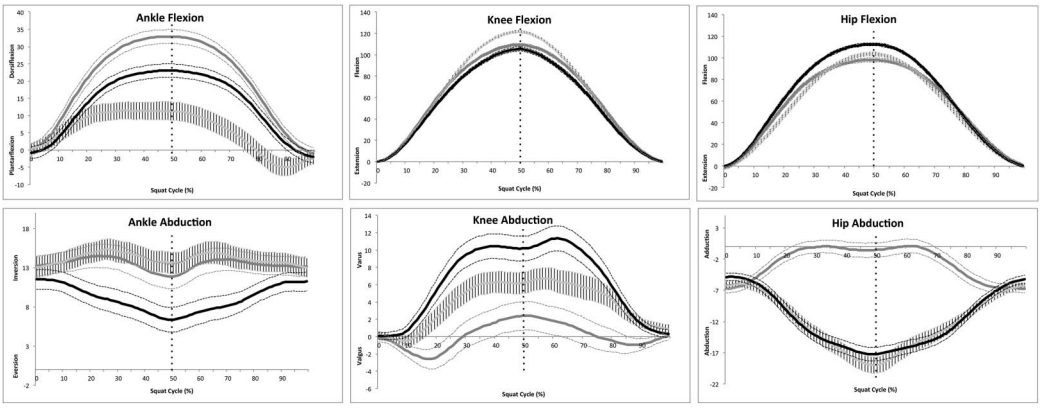


Figure 3. Differences in kinematics during the medio-lateral malaligned squat (grey line), antero-posterior malaligned squat (vertical lines), and control squat (black line) across the squat cycle with 90% confidence intervals. Areas in which confidence intervals did not overlap for 3 or more consecutive points were considered statistically significant.

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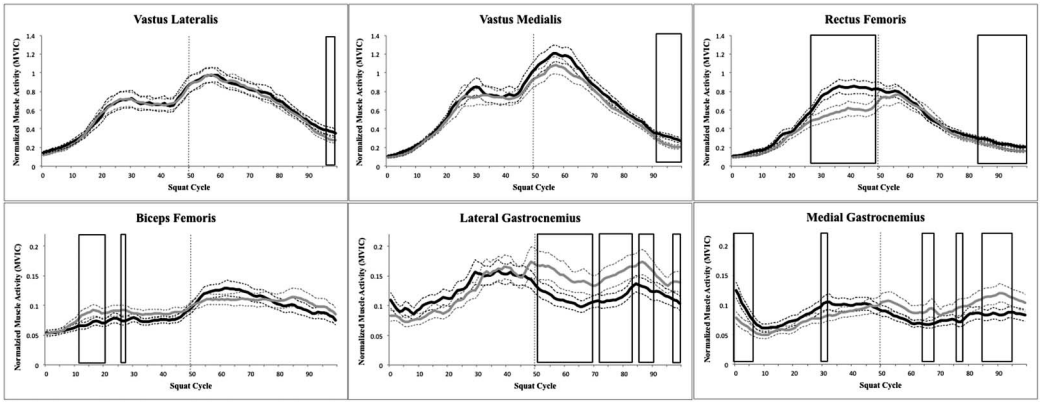


Figure 4. Differences in muscle activation patterns during the medio-lateral malaligned (grey line) and control (black line) squat across the squat cycle with 90% confidence intervals. Areas in which confidence intervals did not overlap for 3 or more consecutive percentage points were considered statistically significant.

Participants also demonstrated significantly decreased hip flexion during 14–71% of the squat cycle compared with the control squat (Figure 3).

Vastus Lateralis. The vastus lateralis had decreased activation during final ascent (96–99%) of the squat cycle in the ML malaligned squat compared with the control squat (Figure 4). Effect size was very large (−6.21) for the significant difference during the squat cycle for ML malalignment (Figure 5).

Vastus Medialis. Vastus medialis activation decreased during the final phase of ascent (92–98%) of the squat cycle in the ML malaligned squat compared with the control squat (Figure 4). Effect size was very large (−3.78) for the difference in activation (Figure 5).

Rectus Femoris. Rectus femoris activation decreased during the initial (15–18%) and final phase of decent (28–48%) of the squat cycle in the ML malaligned squat compared with the control squat. The rectus femoris also displayed

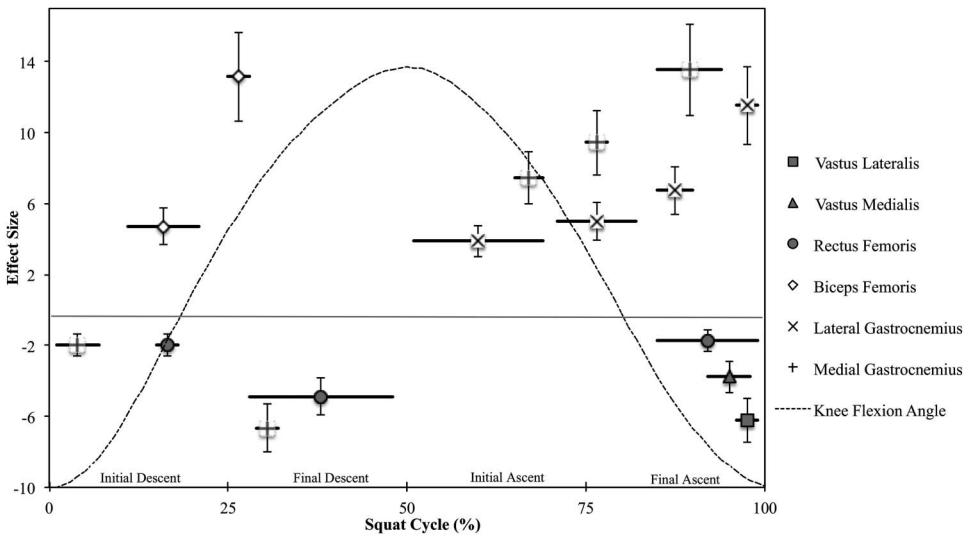


Figure 5. Effect sizes for significant differences between medio-lateral malaligned and control squat. Vertical error bars represent 95% confidence intervals for the effect size point estimate. The horizontal line represents the duration across the squat cycle where confidence intervals did not overlap.

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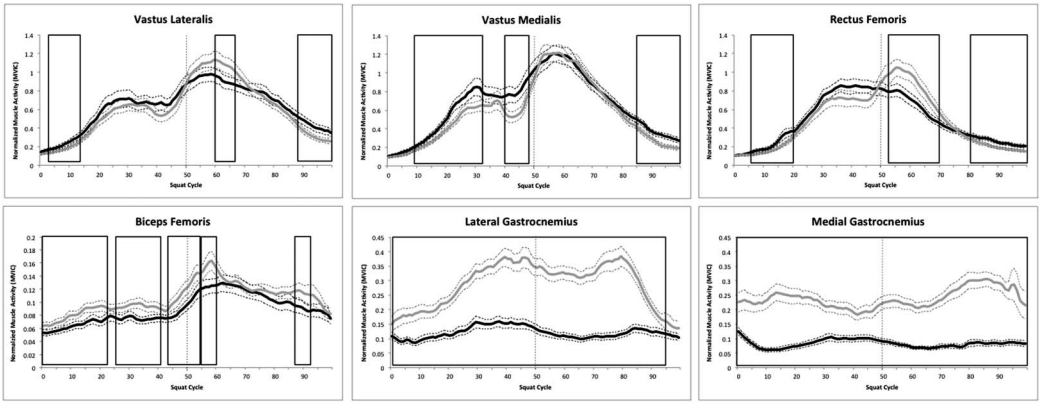


Figure 6. Differences in muscle activation patterns during the antero-posterior malaligned (grey line) and control (black line) squat across the squat cycle with 90% confidence intervals. Areas in which confidence intervals did not overlap for 3 or more consecutive percentage points were considered statistically significant.

decreased activation in the ML malaligned squat during the final phase of ascent (85–99%) of the squat cycle (Figure 4). Effect sizes were very large (Range = −4.90, −1.72) for all differences during the squat cycle (Figure 5).

Biceps Femoris. The biceps femoris activation increased during the initial phase of descent (11–21%) and beginning of the final phase of descent (25–28%) during the ML malaligned squat compared with the control squat (Figure 4). Effect sizes were very large (Range = 4.71, 13.14) for all differences in the ML malaligned squat (Figure 5).

Lateral Gastrocnemius. The lateral head of the gastrocnemius was more active during the ML malaligned squat compared with the control squat in the initial (51–69%) and final phase of ascent (71–82%, 85–90%, 96–99%) during the squat cycle (Figure 4). Effect sizes were very large (Range = 3.90, 11.53) for all differences between the ML malaligned and control squat during the squat cycle (Figure 5).

Medial Gastrocnemius. The medial head of the gastrocnemius was less active during the initial (1–7%) and final phases of

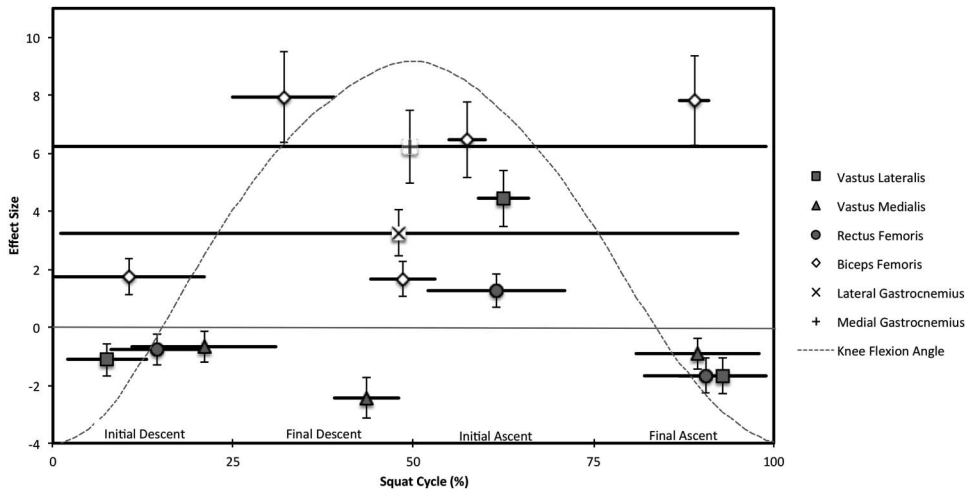


Figure 7. Effect sizes for significant differences between antero-posterior malaligned and control squat. Vertical error bars represent 95% confidence intervals for the effect size point estimate. The horizontal line represents the duration across the squat cycle where confidence intervals did not overlap.

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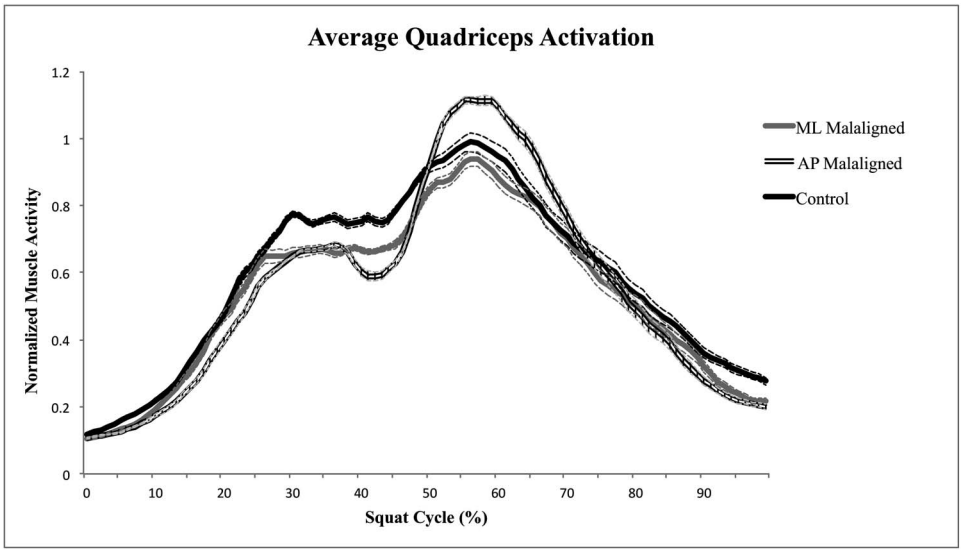


Figure 8. Differences in average quadriceps (vastus lateralis, vastus medialis, and rectus femoris) activation pattern with 90% confidence intervals between squat techniques.

descent (29–32%) of the ML malaligned squat compared with the control squat (Figure 4). During the ascending phases of the squat cycle, the medial gastrocnemius was more active in the ML malaligned squat (65–69%, 75–78%, 85–94%) compared with the control squat (Figure 4). Effect sizes were very large (Range = −1.97, 13.53) for all differences between the ML malaligned and control squat during the squat cycle (Figure 5).

Antero-Posterior Malaligned Squat

Antero-posterior malaligned squats increased anterior knee displacement and decreased lateral knee displacement compared with the control squat (Figure 2). Participants demonstrated significantly less dorsiflexion during the AP malaligned squat during 21–95% of the squat cycle compared with the control squat. The AP malaligned squat increased knee flexion from 22 to 80% of the squat cycle and decreased hip flexion from 5 to 77% of the squat cycle compared with the control squat. Ankle inversion increased from 10 to 92% of the AP malaligned squat compared with the control squat. Participants demonstrated decreased knee adduction during 15–75% of the AP malaligned squat compared with the control squat (Figure 3).

Vastus Lateralis. The vastus lateralis had decreased activation in the AP malaligned squat compared with the control squat during initial descent (2–13%) and final ascent (87–99%) of the squat cycle. Vastus lateralis had increased activation during the AP malaligned squat during initial ascent from peak knee flexion, 59–66% of the squat cycle (Figure 6). Effect sizes were very large (Range = −2.29, 3.47) for all significant differences during the squat cycle for AP malalignment (Figure 7).

Vastus Medialis. The vastus medialis had decreased activation during the initial (11–31%) and final descent (39–48%) of the AP malalignment squat compared with the control squat (Figure 6). Vastus medialis activation also decreased during the final ascent of the squat cycle (81–98%) of the AP malaligned squat compared with the control squat (Figure 6). Effect sizes were moderate to very large (Range = −0.69, −2.44) for all differences during the AP malaligned squat during the squat cycle (Figure 7).

Rectus Femoris. Activation of the rectus femoris decreased during the initial phase of descent (8–21%) and final phase of ascent (82–99%) in the AP malaligned squat compared with the control squat. The rectus femoris activation increased in the AP malaligned squat during the initial phase of ascent (52–71%) (Figure 6). Effect sizes were large to very large (Range = −1.68, 1.26) for all differences during the AP malaligned squat (Figure 7).

Biceps Femoris. The biceps femoris had increased activation in all 4 phases of the AP malaligned squat compared with the control squat (Figure 6). Effect sizes were very large (Range = 1.66, 7.94) for all differences during the AP malaligned squat (Figure 7).

Lateral Gastrocnemius. The lateral gastrocnemius activation also increased during the AP malaligned squat during all phases of descent and ascent (1–95%) compared with the

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control squat (Figure 6). Effect size was very large (3.24) for the difference in activation during the AP malaligned squat (Figure 7).

Medial Gastrocnemius. The medial gastrocnemius was more active during the AP malaligned squat during all phases of descent and ascent (0–99%) compared with the control squat (Figure 6). Effect size was very large (6.24) for the difference in activation during the AP malaligned squat (Figure 7).

Discussion

The main purpose for the inclusion of the body weight squat in training and rehabilitation programs is to increase strength at the thigh, hip, and back musculature (10). The activation patterns of the vastus lateralis, vastus medialis, and rectus femoris during the control squat in this study are similar to those previously reported (8,11,24,28), supporting that the squat exercise focuses on quadriceps activation. The results in this study support the notion that the quadriceps are most active during the concentric phase of the exercise (35,40). The results in this study also support that malaligned squats, both in the sagittal and frontal planes, significantly alters quadriceps activation. The decreased quadriceps activation associated with ML malalignment indicates that frontal plane deviations during a squat alter muscle activation strategy to stabilize the lower extremity during a bilateral squat (Figure 8). Our study agrees with prior findings that the rectus femoris is less active than the vastus medialis and lateralis during a control squat (12); however, frontal plane malalignment further decreased rectus femoris activation during descent into peak knee flexion and increased activation in the knee flexors. The decreased rectus femoris activity during frontal plane malalignment may suggest that increased medial knee displacement during squats changes the nature of the exercise, decreasing quadriceps activation and increasing hamstring and gastrocnemii activity. Further research should continue to investigate the influence of medial knee displacement on rectus femoris activation during closed-chain knee exercises.

In the current study, both AP and ML malaligned squats increased gastrocnemius activation compared with the control squat. The medial and lateral gastrocnemii activation during the descending and ascending phase of the squat was similar to that previously reported during squatting (36). The increased gastrocnemii activation during ML malaligned squats was also similar to increased gastrocnemii activation in individuals with passive medial knee displacement during squatting (36). Participants in this study were instructed to purposefully squat into a malaligned position, which may not represent muscle activation patterns during passive malalignment. The similarities in gastrocnemii activation during passive medial knee displacement indicate that both the medial and lateral gastrocnemii are more active during frontal plane malalignment even with the slight medial knee excursion seen in this study. Increased coactivation of the gastrocnemii during closed kinetic chain exercises stabilizes the ankle during flexed knee stance and decreases the strain at the anterior cruciate ligament by pulling the femur backwards (22,26,34). The increased coactivation of the gastrocnemii during both malaligned squats may indicate an unstable knee joint position with increased anterior and medial knee displacement. These findings support the importance of sagittal plane alignment squat form when patients and clients display even minimal knee abduction especially when the goal of the squat is to strengthen the quadriceps muscle group.

The increased eccentric activation of the knee flexors during malaligned squats may be in an effort to stabilize the knee joint when quadriceps activation decreases and when contact forces are highest. Previous researchers (3,14,15,33,39,41,43) have noted that patellofemoral contact forces are high around 90° of knee flexion, whereas tibiofemoral contact forces are largest when the knee is close to full extension. During both malaligned squats, cocontraction of the biceps femoris and gastrocnemii during parts of the squat cycle when contact forces are highest may be a strategy to stabilize the hip and knee joint (1,8). Hamstring cocontraction during knee flexion also decreases anterior translation and internal rotation, whereas cocontraction of the gastrocnemius decreases strain at the anterior cruciate ligament (16,30), supporting that increased activation of the hamstring and gastrocnemius muscles during malaligned bilateral squats may be a stabilizing technique. Furthermore, the increased activation in the hamstring and gastrocnemii during malaligned squats changes the nature of the exercise, targeting muscles that are considerably less active during a squat with neutral alignment. Further research comparing neutral and malaligned squats should also include gluteus maximus, semitendinosus, and semimembranosus activation. Although gluteus maximus activation reportedly increases with squat depth (4), this may not represent gluteal activation during an unloaded squat to 90° of knee flexion (5) with neutral and malaligned techniques.

In contrast to the decreased quadriceps activation during the ML malaligned squat, the AP malaligned squat increased vastus lateralis and rectus femoris activation during initial ascent. Furthermore, the decreased vastus medialis activation during the AP malaligned squat may be in effort to decrease tibial internal rotation and patellofemoral contact pressure (42). Previous researchers (33) have noted increased patellofemoral contact forces during flexion with increased quadriceps activation, which may lead to the increased eccentric activation of the knee flexors during the AP malaligned squat. Although restricting anterior knee displacement can result in increased thoracic motion and forces at the hip and back during squats (17,27), too much anterior knee displacement may lead to increased patellofemoral contact forces (33,38,39). The knee joint displaced approximately 0.17 m anteriorly compared with neutral

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position during control squats in our study; however, the biceps femoris and gastrocnemius had little activity throughout the squat cycle. Both the ML and AP malaligned squats increased anterior knee displacement by approximately 0.07 and 0.15 m, respectively (Figure 2), which may explain the increase in biceps femoris activation we observed during initial descent and increase gastrocnemius activation during initial and final ascent of the squat cycle (Figures 4 and 6). There is no established "safe zone" for anterior excursion at the knee during squats that can be recommended from the data in the current study. However, we have identified altered muscle activation patterns when alignment is altered during a squat. Further research should explore optimal anterior knee displacement during bilateral squatting to ensure that the spine, hip, and knee are not exposed to risk during the exercise.

There were some limitations to this study including the lack of standardization of knee flexion angle, squat velocity, and reliability of EMG findings. Although knee flexion angle was not standardized, all participants received the same verbal instructions and these instructions were interpreted in a similar manner given the tight confidence intervals. Squat velocity was not standardized; however, both the descending and ascending phases of the squat were reduced to 50 points in order to standardize each squat based on kinematic events. Future research using this technique should standardize squat velocity to further minimize changes in muscle activity. Although we did not assess reliability of EMG in the current study, reliability of surface EMG using a repeated measures design has been well documented during functional tasks in both healthy and pathological populations (23,25,31). Lastly, the order of squat performance was not counterbalanced, with the control condition performed last. This was an active decision to limit any feedback during squat performance until malaligned squats were completed. Participants were also given adequate rest between squats, limiting the influence of the previous squat. Future researchers using this design should consider counterbalancing the order of the malaligned squat technique to further limit the influence of one squat variation on another.

The results of this study support that malaligned squats in the frontal and sagittal plane significantly alter muscle activation patterns in the lower extremity, increasing activation in hamstring and gastrocnemius muscles compared with a control squat. Frontal and sagittal plane knee excursion also significantly alter quadriceps activation patterns during squatting, changing the demands of the task on the knee musculature. Despite the altered activation strategies during malaligned squats, activation in the hamstring and gastrocnemius decreased during the control squats using basic instructions and feedback. The simple cues used in this study may help guide clients and patients to activation in the quadriceps and decrease activation in the hamstring and gastrocnemius during bilateral bodyweight squats.

Practical Applications

The bilateral squat exercise is a commonly used exercise for strengthening the quadriceps. Oftentimes, the exercise is not executed properly without initial instruction from a practitioner. Two common malalignments during a bodyweight bilateral squat are medial and anterior knee displacement; however, there is little information about the changes in muscle activation patterns resulting from these malalignments. The results in this study support that medio-lateral and antero-posterior malalignments alter muscle activation patterns in the lower extremity, specifically increasing activation of the hamstrings and gastrocnemii, which have relatively low activity in a neutrally aligned squat. Increased cocontraction of the knee flexors and gastrocnemii during malaligned squats may be in an effort to stabilize the ankle, knee, and hip during flexed knee stance, indicating that malaligned knee positions may be potentially injurious. The increased quadriceps activation with increased anterior knee excursion around peak knee flexion should also be a consideration in strength and conditioning programs and inclusion of squats similar to the ballet plié squat should be cautioned. Furthermore, the results of this study support the use of the bilateral squat as an assessment tool for clients and patients who complain about tightness and pain in the hamstring or gastrocnemii.

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